

A Short Guide to Lemma C.5

Translation-covariant scalar kernels are convolutions

Abstract

Lemma C.5 identifies every positive translation-covariant map from $L^1(\mathbf{H})$ to finite measures. Such a map must first dilate the input location by $\sqrt{\gamma}$ and then add noise with a fixed finite measure μ . The proof obtains μ by feeding the map an approximate point mass at the origin. This note explains the covariance convention, the weak-* extraction, and the passage from approximate point masses to arbitrary L^1 inputs.

1 The statement in probabilistic language

Suppose $u(x) dx$ is an input density on the finite-dimensional vector space $\mathbf{H}$. A positive map $\mathcal{R}$ sends it to a finite output measure. The covariance assumption says that translating the input by h translates the output by $\sqrt{\gamma}h$:

$$\mathcal{R}(u(\cdot - h))(E) = \mathcal{R}(u)(E - \sqrt{\gamma}h). \quad (1)$$

Lemma C.5 says that there is a single finite positive measure μ such that

$$\mathcal{R}(u)(E) = \int_{\mathbf{H}} u(x) \mu(E - \sqrt{\gamma}x) dx. \quad (2)$$

Thus, conditionally on an input point x , the output is

$$Y = \sqrt{\gamma}x + R, \quad R \sim \mu.$$

The same additive noise law is used at every x ; translation covariance forbids any location-dependent noise.

The measure μ may have total mass below one. In the application, $\mathcal{R}$ is obtained as a weak-* limit of classical trace measures, so some probability may escape to infinity.

2 Functional-analytic setting

The output space is

$$C_0(\mathbf{H})^*,$$

which the Riesz representation theorem identifies with the finite Radon measures on $\mathbf{H}$. Weak-* convergence of measures means

$$\int f d\mu_j \longrightarrow \int f d\mu \quad (f \in C_0(\mathbf{H})).$$

The map

$$\mathcal{R} : L^1(\mathbf{H}) \longrightarrow C_0(\mathbf{H})^*$$

is a positive contraction. Hence, for $u \geq 0$,

$$\|\mathcal{R}(u)\| \leq \|u\|_1.$$

In particular, normalized nonnegative inputs produce measures of mass at most one.

3 Statement of Lemma C.5

Lemma C.5 (rephrased). *Let $\mathcal{R} : L^1(\mathbf{H}) \rightarrow C_0(\mathbf{H})^*$ be a positive contraction satisfying (1). Then a finite positive measure μ with $\mu(\mathbf{H}) \leq 1$ exists such that (2) holds for every $u \in L^1(\mathbf{H})$ and every Borel set E .*

4 Proof walkthrough

Step 1: probe the kernel at the origin

Choose a nonnegative approximate identity $u_j \in C_c(\mathbf{H})$:

$$\int_{\mathbf{H}} u_j(x) \, dx = 1, \quad \text{supp}(u_j) \longrightarrow \{0\}.$$

One may think of u_j as an increasingly concentrated approximation to the point mass δ_0 . If a genuine transition kernel existed, its output on δ_0 would be the noise measure. Accordingly, consider

$$\mu_j := \mathcal{R}(u_j).$$

Positivity and contractivity imply $\mu_j(\mathbf{H}) \leq 1$.

The unit ball of $C_0(\mathbf{H})^*$ is weak-* compact. Since $C_0(\mathbf{H})$ is separable, a subsequence converges weak-* to a positive measure μ :

$$\int f \, d\mu_j \longrightarrow \int f \, d\mu \quad (f \in C_0(\mathbf{H})). \quad (3)$$

The norm is weak-* lower semicontinuous, so $\mu(\mathbf{H}) \leq 1$. Equality need not hold because the constant function 1 does not belong to $C_0(\mathbf{H})$ when $\mathbf{H}$ is noncompact.

Step 2: smooth an arbitrary input

Use the convolution convention

$$(u * u_j)(y) = \int_{\mathbf{H}} u(x) u_j(y - x) \, dx = \int_{\mathbf{H}} u(x) u_j(\cdot - x)(y) \, dx.$$

The approximate-identity property gives

$$u * u_j \longrightarrow u \quad \text{in } L^1(\mathbf{H}). \quad (4)$$

Bounded linearity of $\mathcal{R}$ therefore implies

$$\mathcal{R}(u * u_j) \longrightarrow \mathcal{R}(u)$$

in the norm of finite measures.

On the other hand, covariance applied to each translated copy of u_j gives

$$\begin{aligned}\mathcal{R}(u * u_j)(E) &= \int_{\mathbf{H}} u(x) \mathcal{R}(u_j(\cdot - x))(E) \, dx \\ &= \int_{\mathbf{H}} u(x) \mathcal{R}(u_j)(E - \sqrt{\gamma}x) \, dx.\end{aligned}\tag{5}$$

Thus the desired convolution formula is already true with μ_j in place of μ , but for the smoothed input $u * u_j$.

Step 3: pass to the weak-* limit

To justify the limit in (5), pair both sides with an arbitrary $f \in C_0(\mathbf{H})$. The right side becomes

$$\int_{\mathbf{H}} u(x) \left(\int_{\mathbf{H}} f(r + \sqrt{\gamma}x) \mu_j(\,dr) \right) \, dx.\tag{6}$$

For each fixed x , the translated function $r \mapsto f(r + \sqrt{\gamma}x)$ still belongs to $C_0(\mathbf{H})$, so (3) gives convergence of the inner integral to the same expression with μ . Its absolute value is bounded by $\|f\|_{\infty}$, and $|u|$ is integrable. Dominated convergence therefore turns (6) into

$$\int_{\mathbf{H}} u(x) \left(\int_{\mathbf{H}} f(r + \sqrt{\gamma}x) \mu(\,dr) \right) \, dx.$$

The left side converges to the pairing of f with $\mathcal{R}(u)$ by (4). Since $C_0(\mathbf{H})$ separates finite Radon measures, (2) follows.

5 Examples

1. If $\mu = \delta_0$, then

$$\mathcal{R}(u)(E) = \int u(x) \mathbf{1}_E(\sqrt{\gamma}x) \, dx,$$

which is the deterministic dilation $x \mapsto \sqrt{\gamma}x$.

2. If μ is a probability law, the map performs the dilation and adds independent noise distributed as μ .
3. If $\mu(\mathbf{H}) = c < 1$, the same procedure succeeds with the input-independent probability c .

No density for μ is required. The noise may have atoms or a singular part, which is important because weak-* limits of absolutely continuous output measures need not remain absolutely continuous.

6 Role in Proposition C.6

In the hybrid converse, $\mathcal{R}$ is the limiting classical trace kernel of Følner-averaged hybrid channels. Lemma C.5 isolates all its freedom in the one measure μ . The classical Gaussian witness then yields

$$\int c_{\gamma}(r) \mu(\,dr) \leq B_{\text{cl}}(\gamma, d) \mu(\mathbf{H}) \leq B_{\text{cl}}(\gamma, d).$$

That bound controls the trace of the weighted quantum limit and supplies the classical factor in the final hybrid fidelity bound.